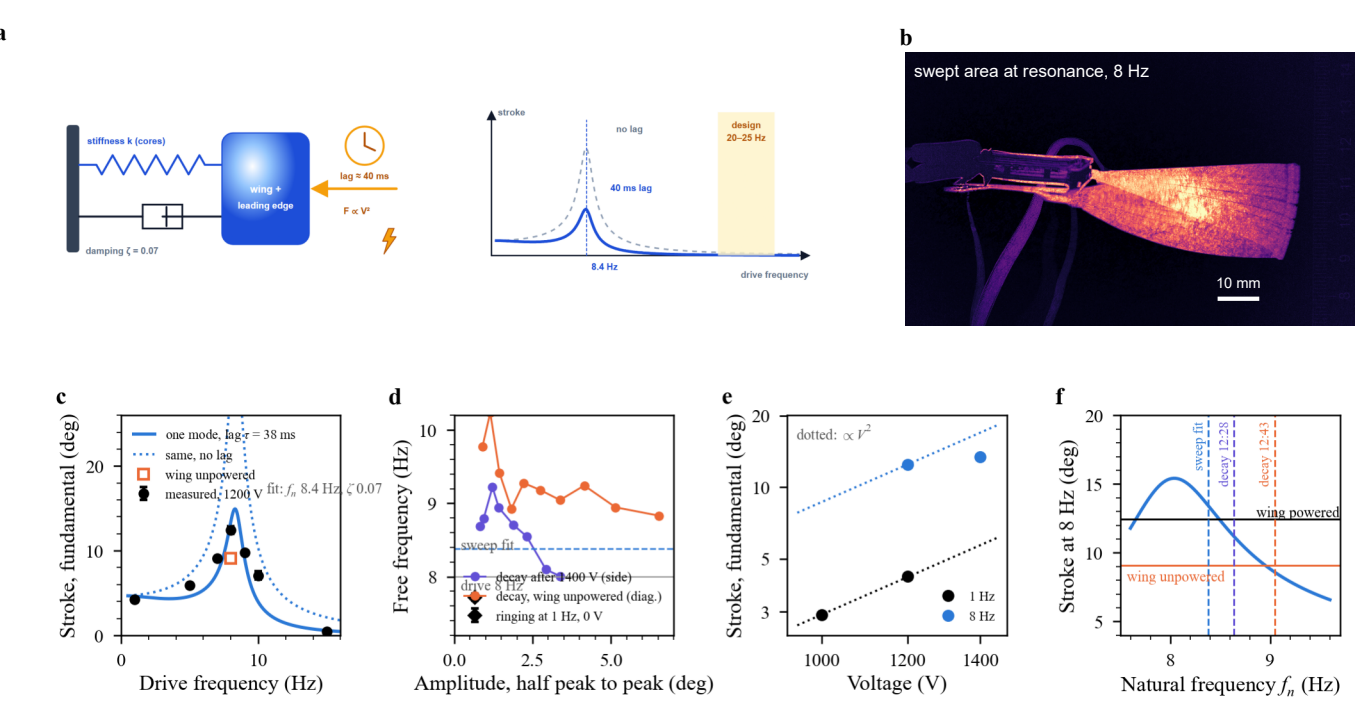

a
stiffness k (cores)
wing + leading edge
lag ≈ 40 ms
F ∝ V²
damping ζ = 0.07
stroke
no lag
40 ms lag
design 20–25 Hz
8.4 Hz
drive frequency
b
swept area at resonance, 8 Hz
10 mm
c
one mode, lag τ = 38 ms
same, no lag
wing unpowered
measured, 1200 V
fit: f_n 8.4 Hz, ζ 0.07
Stroke, fundamental (deg)
Drive frequency (Hz)
d
sweep fit
drive 8 Hz
decay after 1400 V (side)
decay, wing unpowered (diag.)
ringing at 1 Hz, 0 V
Free frequency (Hz)
Amplitude, half peak to peak (deg)
e
dotted: ∝ V^2
1 Hz
8 Hz
Stroke, fundamental (deg)
Voltage (V)
f
sweep fit
decay 12:28
decay 12:43
wing powered
wing unpowered
Stroke at 8 Hz (deg)
Natural frequency f_n (Hz)